



TRAINING PROGRAMME

Zero Trust & Cloud Security

Plan a zero trust adoption and protect data and AI systems in the cloud

15 HOURS

FOR Security managers, architects and IT leaders planning a security strategy



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Overview

This workshop helps security and IT leaders plan a move to zero trust and protect data in the cloud. Participants learn the principles of NIST SP 800-207 and the CISA Zero Trust Maturity Model, with identity at the centre, then work on cloud data security: encryption and key management, data loss prevention and data residency. They see how AI changes detection and response and how to secure AI systems themselves. The workshop closes with a final project in which each team builds a zero trust adoption roadmap for a given organisation.

At a glance

#	PHASE	FOCUS	HOURS
01	Zero Trust	NIST SP 800-207, CISA Zero Trust Maturity Model, identity-centric security	3 h
02	Cloud Data & AI Security	Encryption and keys, DLP, data residency, AI for detection, securing AI systems	6 h
03	Final Project	A zero trust adoption roadmap, presented to the group	3 h
	Buffer	Catch-up, revision, extra lab time	3 h
Total			15 h

Key facts

AUDIENCE	Security managers, architects and IT leaders planning a security strategy
PREREQUISITES	None. A general IT or security background is expected (course: Cybersecurity for IT Teams, recommended)
DURATION	15 h: 12 h of chapters + 3 h buffer
FORMAT	Workshop: short lectures, each followed by group exercises on an organisation case
LAB ENVIRONMENT	One laptop per participant with a browser; maturity assessment templates; trainer demonstrations in the AWS console (KMS, Macie); no account or cost for participants
ASSESSMENT	An exercise validated at the end of each phase and a final roadmap presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can assess an organisation against the CISA Zero Trust Maturity Model and build a prioritised roadmap covering identity, data and AI systems

PHASE 01

3 H

Zero Trust

Participants learn what zero trust means in practice: the tenets and architecture of NIST SP 800-207, the five pillars, three cross-cutting capabilities and four maturity stages of CISA ZTMM v2.0, and why identity becomes the main control. Validated exercise: the case organisation rated on the five CISA pillars.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Zero Trust Principles	NIST SP 800-207, CISA Zero Trust Maturity Model, identity-centric security	Lecture + Workshop	3 h

PHASE 02

6 H

Cloud Data & AI Security

Participants decide how cloud data is encrypted and who holds the keys, where data loss prevention fits and how residency rules constrain the design, then look at AI on both sides: as a help for detection and response, and as a new system to secure. Validated exercise: a data protection plan for the case organisation's most sensitive data set.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Cloud Data Security	Encryption, key management, DLP, data residency	Lecture + Workshop	3 h
2.2	AI in Security	AI-enabled detection and response, securing AI systems	Lecture + Workshop	3 h

PHASE 03

3 H

Final Project

Teams of two or three build a zero trust adoption roadmap for a given organisation, reusing the workshop exercises, and present it as they would to an executive committee.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Assess & Plan	Current maturity, target state, prioritised initiatives and milestones	Project	1.5 h
3.2	Present & Review	Present the roadmap; questions and feedback	Project	1.5 h

FINAL PROJECT

Zero Trust Adoption Roadmap

A zero trust roadmap for a given organisation: current maturity on the CISA pillars, target state, prioritised initiatives for identity, data and AI systems, and milestones.

STACK NIST SP 800-207, CISA Zero Trust Maturity Model, assessment templates

DELIVERABLE Maturity assessment, roadmap, presentation